QueryBook — Drawings and Description of the Drawings

FIG. 1–61 · companion to the Provisional Patent Application (synchronized, with support for agent-wrapper services, the self-knowledge layer, the digital synaptic network with network memory, and the four-state Fact Unit)

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Brief Description of the Drawings

The accompanying drawings, incorporated in and forming part of the specification, illustrate embodiments of the invention and, together with the description, serve to explain its principles. In the drawings, like reference numerals designate corresponding parts throughout the several views.

Platform (FIG. 1–18)

FIG. 1 is a block diagram of the system, showing the ingestion pipeline, the storage engine, the UFCS substrate, the FQL query engine, the verification ledger, and the federation boundary.

FIG. 2 is a diagram of a Fact Atom carried within a Fact Envelope, showing the nucleus, group classification, polarity, and links of the atom, together with the envelope's confidence, provenance, certification, and content-derived identifier.

FIG. 3 is a diagram of a fact-certification authority hierarchy, showing a root trust anchor, domain authorities, and the certified facts and trust scores derived beneath them.

FIG. 4 is a diagram of the substrate relationships, showing a pairwise graph, a hyperedge, and a directed hyperchain among Fact Units.

FIG. 5 is a diagram supporting claim 11, showing tour/exhibition sequencing of immersive scenes into ordered waypoints.

FIG. 6 is a diagram of a query that filters jointly on fact structure and certification trust, and of the trust-native execution by propagation, branch-and-bound pruning, and contradiction exclusion.

FIG. 7 is a flowchart of verifying machine-generated text against certified records, classifying each assertion as verified, contradicted, or unknown.

FIG. 8 is a diagram of federation, showing a certified envelope crossing a boundary and being re-validated on the receiving side.

FIG. 9 is a diagram of fact-level erasure using an externally held key (cryptographic shredding of a sealed payload).

FIG. 10 is a diagram of the Universal Security Architecture, showing the twelve defensive layers, the five simultaneous defensive postures, and key mechanisms including the AI Attack Prediction Engine.

FIG. 11 is a diagram of Universal Honeypot Saturation and the Data Firehose, showing port saturation via the Phantom Port Daemon and application-entry-point saturation via the Shadow-App mechanism.

FIG. 12 is a diagram of the five deployment topologies — cloud SaaS, local server, hybrid, single-device, and edge — over the invariant common layer of safety, governance, and provenance.

FIG. 13 is a diagram of the Adaptive Response Engine, showing the five-stage pipeline from Result-Set Analysis through Generation-Directive Assembly, with the directive fixed before generation begins.

FIG. 14 is a diagram of the user-owned agent system, showing Blueprint definition and registration, Agent-Instance creation under commissioner-bounded authority, and the governance evaluation at each action.

FIG. 15 is a diagram of deterministic multi-language regeneration, showing source ingestion to Fact Units, the language-independent Fact Unit graph as a single knowledge source, and regeneration into any target language.

FIG. 16 is a diagram of semantic compression, expansion, colorization, and resolution upscaling, showing the meaning-storage compression flow and the on-demand regeneration of a surface representation.

FIG. 17 is a diagram of the QueryBook-enhanced TCP/IP hybrid protocol packet, showing the optional semantic-metadata header carrying FUID, ProvenanceRef, and confidence fields alongside the standard headers and payload.

FIG. 18 is a diagram of the Neuromorphic Cognitive Layer and attractor network, showing the energy function, the capacity ceiling, and convergence to a stable equilibrium.

Confidence, packet, framing and query (FIG. 19–22)

FIG. 19 is a diagram of the Fact Unit confidence and evidence model, comprising an evidence pair, a Beta posterior, a score, a credible interval, source diversity, temporal decay, and a confidence trend.

FIG. 20 is a diagram of the UFCS packet structure, showing the fields of a serialized Fact Unit record and an exemplar packet.

FIG. 21 is a diagram of UFCS packet framing, in which immutable packets are de-duplicated by fingerprint and appended to content-addressed blocks under a fingerprint index.

FIG. 22 is a diagram of FQL query anatomy with a worked example, in which match, where, exclude, rank, and return clauses produce a highest-trust answer bound to its evidence by a provenance hash.

Language and speech subsystem (FIG. 23–30)

FIG. 23 is a block diagram of the overall QueryBook Language Expression Layer (LEL) system architecture.

FIG. 24 is a block diagram of the Sub-Language Priming Layer (SLPL) and the measurable representation produced by each of its six components.

FIG. 25 is a flow diagram of the developmental learning sequence and the one-way Transition Gate having four readiness thresholds.

FIG. 26 is a block diagram of the multilingual Delta Acquisition Model and the acquisition acceleration flywheel.

FIG. 27 is a block diagram of the seven-stage Speech Output Pipeline.

FIG. 28 is a data-flow diagram of the co-grounded integration of the LEL with the Federated Query Language (FQL) and the Universal Federated Connectivity System (UFCS).

FIG. 29 is a block diagram of a reduced-to-practice embodiment (the Language Lab) implementing Phase One, the Transition Gate, and a deterministic Phase Two.

FIG. 30 is a data-flow diagram of the deterministic Phase-Two semantic grounding of the reduced-to-practice embodiment, in which word meanings are attached only from auditable sources — a bundled public-domain dictionary and self-grounding against verified Fact Units in the store — with any large-language-model meaning source excluded from the grounding path.

Further subsystems and embodiments (FIG. 31–49)

FIG. 31 is a block diagram of the advanced multilingual pronunciation subsystem and self-contained speech output of the reduced-to-practice embodiment, showing the pronunciation preference ladder (bundled open-source phonemizer, then per-language rule-seeded grapheme-to-phoneme tables, then an explicit non-fabricated null), the single-call phrase transcription that keeps the displayed transcription consistent with the synthesized audio, and the self-contained bundle and engine-selection ladder that produce offline audio or an honest refusal without fabricating audio.

FIG. 32 is a flow diagram of the permanent, deterministic pronunciation-and-speech quality-control battery of the reduced-to-practice embodiment, which, for each supported language and without a large language model, verifies that a phonemic transcription is produced, that the syllable count is within tolerance, and that a short phrase synthesizes to a valid audio waveform, returning a per-language report and an overall pass score.

FIG. 33 is a data-flow diagram of the Language & Identity Linking (LIL) voiceprint enrollment and matching subsystem, in which a consent-gated capture produces an acoustic speaker embedding (voiceprint), a match against stored voiceprints is passed through a QueryBook trust gate returning VERIFIED, CONTRADICTED, or UNKNOWN, and verified identity links are written as provenance-tracked, trust-scored Fact Units in the identity graph, with cross-script name and entity resolution performed by the language subsystem.

FIG. 34 is a flow diagram of the LIL trust-gated voice authentication process, in which a claimed-identity verification produces a match score and anti-spoofing signals that feed a deterministic risk-scoring policy and a decision gate returning accept, challenge (second factor), or reject, each decision recorded as a provenance-tracked auth-event Fact Unit.

FIG. 35 is a block diagram of the Universal Integrity Audit Service (UIAS), in which connector and ingestion agents observe an unmodified monitored system at multiple workflow positions and seal each observable event as a content-addressed Fact Unit carrying a content hash, a one-way submitter-identity hash, an independent nanosecond timestamp, a source fingerprint, and the sealing agent's identity; a similarity-detection agent sets a contemporaneous cross-party anomaly flag over content signatures without storing content; and a verification agent continuously re-verifies the append-only Fact-Unit ledger.

FIG. 36 is a flow diagram of the UIAS signed Proof Package generation, in which, on an accusation or inquiry, a proof-package agent assembles verified sealed Fact Units, the similarity-log entry, a chronological ledger extract bound by the response-provenance hash, and a factual determination from the arbitration agent under the Prime-Directive gate, and signs the package under the UIAS independent identity to produce independent, pre-dispute, tamper-proof, complete, and admissible evidence.

FIG. 37 is a data-flow diagram of the Paralinguistic & Pragmatic Interpretation Layer (PIL) applied to an ingested voice, in which a paralinguistic analyzer produces measured acoustic readings and a labeled arousal estimate while a pragmatic framer applies sourced context, perspective, ergonomics, culture, and social-mores frames, the two combining into an Interpretation Frame written as provenance-tracked Fact Units, with estimates labeled and never asserted as true feelings.

FIG. 38 is a flow diagram of the PIL human-query interpretation, in which a query is reduced to an Interpretation Frame, intent is disambiguated from context and perspective, an ambiguity gate asks one clarifying question when intent is materially uncertain, a register-and-ergonomic-mode selector sets the reply style and delivery from sourced culture/mores/ergonomics packs, and the existing retrieve-gate-compose pipeline answers from verified Fact Units in that register without altering the facts.

FIG. 39 is a block diagram of the Lecture Query mode ingestion pipeline, in which a live or recorded audio/video program is captured under consent, transcribed and diarized, analyzed by the paralinguistic layer over time and by the language layer for structure and claims, summarized deterministically, and ingested as a Lecture Record whose claims are stored as attributed utterance facts and which is forever-linked to the speaker, venue, event, organization, and date by immutable provenance-tracked, trust-gated identity links.

FIG. 40 is a data-flow diagram of the forever-linked Lecture Record in the Fact-Unit identity graph and its querying, showing the record bound to speaker, venue, event, and organization entities and to the transcript and paralinguistic track, its attributed claims gated as corroborated, contradicted, or open, and a query answered with citations to the record and attribution to the speaker, refused where unknown.

FIG. 41 is a data-flow diagram of deterministic multilingual acquisition without a generative model, in which a per-language learning agent acquires vocabulary, phonology (IPA), and grammatical regularities from a lawful public-domain corpus and a governed bilingual dictionary, an IPA transcription is verified against a reference phonemizer and withheld rather than asserted when verification fails, and each accepted item is written as a provenance-tracked lexical record reproducible from the same inputs, with no large or generative language model in the grounding path.

FIG. 42 is a block diagram of a Dialect Parameter Cluster (DPC), a declared and versioned cluster of phonological, lexical, and orthographic parameters by which an input utterance is attributed to a dialect by a deterministic parameter-match score, the cluster modulating expression only and never altering admission, applicability scope, confidence, or the governed vocabulary.

FIG. 43 is a data-flow diagram of grounded scene compilation to an external generative media engine, in which a grounded scene representation derived from verified Fact Units is compiled and validated into a prompt bundle for an external generative video engine and any returned media is admitted only as a derived artifact marked as a reconstruction and never as a grounding source.

FIG. 44 is a block diagram of key-safe external-service mediation, in which a credential for an external service is held on the server side of the query/presentation boundary and never transmitted to the client, the server performing the external call and serving the result to the client by an opaque cache identifier so that the credential never crosses the boundary.

FIG. 45 is a flow diagram of scene reconstruction from prose, in which a deterministic reconstructor derives a shot sequence, a timed prompt sequence, and pacing and music cues as a bounded reconstruction, renders a participant who is not an identified subject as a silhouette so that no unverified identity is fabricated, and marks the whole as a reconstruction.

FIG. 46 is a block diagram of modular edition packaging, in which the system is deployable as one of a plurality of nested capability editions selected at launch, each edition enabling a declared subset of subsystems while the Prime Directive and the deterministic reasoning core remain invariant across every edition.

FIG. 47 is a block diagram of the version-independent, provenance-continuous store, in which the knowledge store resides at a canonical location independent of the deployment software version so that successive software versions bind to the same provenance-tracked records, a version change never re-initializes the store, and the system refuses to substitute an empty store where a populated store exists.

FIG. 48 is a data-flow diagram of a cited question-and-answer (ChatBot) application embodiment, in which a user question is interpreted for intent and register, answered by Federated Query Language retrieval over verified Fact Units, passed through the Prime-Directive gate, and returned either as a composed answer carrying per-fact citations or, where the evidence is insufficient, as an honest UNKNOWN, the same citation-or-UNKNOWN contract holding when the embodiment is deployed as a ChatBot or assistant surface.

FIG. 49 is a block diagram of the application and deployment surfaces of the system over its invariant core, in which a single governed body of Fact Units is presented through an on-device micro-encapsulated eBook or document question-and-answer surface, a software-as-a-service or application-programming-interface service, a Fact-Unit compressed search engine, a deterministic large-language-model-hybrid adjunct, and a private or offline analysis deployment, each surface inheriting the same citation-or-UNKNOWN contract and provenance and none able to assert a fact the core would refuse.

Agent-era extensions (FIG. 50–55)

FIG. 50 is a block diagram of Open Claw ingestion, showing content-based format detection, archive bounds, the Artifact Segmenter, the instruction-like content screen, the OpenClaw Interpreter, and the admission of artifact-description, structured, passage, and flagged untrusted records to the Fact Unit store under a provenance seal.

FIG. 51 is a block diagram of on-chain certification, showing a QueryBook object and its local provenance seal, the content hash and on-chain metadata, the in-system transaction signer and network guard, the certificate contract on a public blockchain, the independent verifier, and the licence check.

FIG. 52 is a block diagram of the agent protection and deception layer, showing the HTTP gate, the isolation register, the Agent Gateway, the deception check with honeycode, honeytokens, tripwires, and honeypot agents, the AI-aware firewall and anomaly detector, the security event bus, the decoy environment, the engagement recorder, and operator-only release.

FIG. 53 is a block diagram of the tamper-evident security record, showing append-only entries, the hash chain, write-once signed Merkle checkpoints, a mirror on a second device, a blockchain anchor, the recompute-from-content verifier with inclusion proofs, and the assurance report.

FIG. 54 is a block diagram of agent interoperability, showing an external AI agent and its skill definition, the Model Context Protocol tool server, the tool-to-command mapping, the Agent Gateway with the agent’s allow-list, grounded handlers, and isolation upon a reserved command.

FIG. 55 is a block diagram of the framed hybrid transport, extending FIG. 17, showing per-type compression, the frame with its header, metadata, payload, CRC32 checksum, and optional signature, the control, normal, and bulk lanes, the receiver, package rebuild with manifest check, and record verification into the Fact Unit store.

Support for agent-wrapper services (FIG. 56)

FIG. 56 is a block diagram of the support interfaces QueryBook provides to external agent-wrapper services such as Verified Agent Transport, grouped as verification, transport, record and certificate, and governance interfaces over the Fact Unit store, security log, certificates, Agent Gateway and MCP server, and framed hybrid transport.

Self-knowledge (FIG. 57)

FIG. 57 is a block diagram of the self-knowledge layer, showing a self-question resolved through a specification index, an evidence-checked implementation map over the program's own source files and self-test results, a status labeller, an aspect describer, a consistency checker and gap agenda, a rule-based answerer with a disclosure filter, and the interfaces through which the answer, with its label and sources, is returned.

Digital Synaptic Network — answer routing (FIG. 58)

FIG. 58 is a block diagram of the digital synaptic network's answer routing, showing a lookup entering an answer router with a central, shadow or routed setting, a pathway table of learned store pathways, a complete-answer fallback ordered by spreading activation, a shadow comparison log and health-report verdict, a re-check of pathways on re-enable, and a stress-test harness.

Network memory (FIG. 59)

FIG. 59 is a block diagram of the network memory, showing activity sources feeding a synapse store of digital synapse records, a plasticity unit, a turnover unit with consolidation, a co-activation unit, an engram store, an assembly detector, a detector of links between assemblies, a spreading-activation unit and the interfaces through which the network is inspected.

Four-state Fact Unit (FIG. 60)

FIG. 60 is a block diagram of the four-state Fact Unit, showing a stored fact entering a rating unit with a backing selector, a corroboration rule, a state table of four truth states and a weight table, the rated fact and tally returned, a two-bit evidence encoder, a join unit operating by bitwise OR over packed states, and a hardware mapping to binary, four-level and higher-level devices.

Fact Unit identity state and context (FIG. 61)

FIG. 61 is a block diagram of the identity state and the context of a Fact Unit, showing an identity unit with four identity states, an as-printed fact and an unresolved zone, a mention weigher, a context table held beside the facts, a fill unit reading the cited source, and the separation of measurements and inferences from facts.

Detailed Description of FIG. 56

FIG. 56 shows the interfaces QueryBook provides to external agent-wrapper services — Verified Agent Transport (VAT) and similar relays that sit between an AI agent and the network and check what the agent says and does. The wrapper service itself is outside the system and is described separately; FIG. 56 shows only what QueryBook offers it.

The wrapper service (5690) receives the traffic of an AI agent (5695) bound for a destination (5697) and calls the support interfaces (5600) under an identity registered with the Agent Gateway. Verification is read-only and records are append-only: a wrapper service never writes Fact Units.

Reference numerals

Numeral	Element	Rests on
5600	QueryBook support interfaces	
5610	Verification interfaces	FIG. 6, 7, 21, 22
5612	Claim verification endpoint (read-only): candidate Fact Units or fingerprints in; verified, contradicted or unknown out, with citation and trust	FIG. 22
5614	Contradiction query template: same subject and predicate, conflicting value or opposite polarity, trust at or above the minimum	FIG. 6, 7
5616	Fingerprint-only verification mode: fingerprints and canonical subject/predicate identifiers only	FIG. 21
5618	Triple extraction and assertion-type labels (fact, opinion, plan, prediction)	FIG. 2, 6
5620	Transport interfaces	FIG. 55
5622	Frame message types 16–31 reserved for registered extensions; unregistered types dropped and recorded	FIG. 55
5624	Extension flag (0x0040) and reserved header field as 16-bit session number and 16-bit hop count	FIG. 55
5626	Python frame codec; control, normal and bulk lanes as QUIC streams with TLS-on-TCP fallback	FIG. 55
5628	Have/want fingerprint reconciliation; Bloom filter above 4,096 fingerprints	FIG. 21, 55
5630	Record and certificate interfaces	FIG. 51, 53
5632	Signed verdict records and published verifier key	FIG. 53
5634	Append by a registered service to the security log	FIG. 53
5636	Session-range Merkle root and inclusion proofs	FIG. 53
5638	Third-party session certificate type, sealed as a Fact Unit	FIG. 51
5640	Governance interfaces	FIG. 52, 54
5642	Wrapper service registration (read-only verification, append-only record rights)	FIG. 52

Numeral	Element	Rests on
5644	Signed, versioned, pull-only policy bundle export	FIG. 52
5646	Tool-server pre-call and post-call hooks; hash of each tool result record	FIG. 54
5648	Principal evidence pair (agent reputation); restriction only on a recorded operator decision	FIG. 19, 52
5650	Fact Unit store and FQL	FIG. 2, 6, 21, 22
5660	Security log	FIG. 53
5670	Certificates	FIG. 51
5680	Agent Gateway and MCP server	FIG. 52, 54
5685	Framed hybrid transport	FIG. 55
5690	External agent-wrapper service (e.g., Verified Agent Transport) — outside the system	
5695	AI agent	
5697	Destination (agent, tool or data source)	

Status

The support interfaces of FIG. 56 are specified and not yet implemented; no performance figures are asserted for them. They correspond to features 364–382 of the QueryBook Composite Feature Registry v68 and to claims 355–357 of the application.

Detailed Description of FIG. 57

The Self-Knowledge Layer (SKL) is the part of QueryBook that describes QueryBook. Asked what it is, how it reasons, whether it is neuromorphic, whether it predicts, or whether it improves itself, the system answers from three verified layers and from nothing else: the specification (the numbered sections of this Overview), the implementation (the module and symbol that carry out each section, checked against the program's own source files at the moment of asking), and the evidence (the host application's own self-tests, diagnostics and measured scores). The layer never generates its self-description with a language model, and it never repeats a claim its own code does not support.

A self-question (5700) from a user, the assistant or an application is received by the self-knowledge layer (5705). A specification index (5710) holds the numbered sections of the specification and resolves the question to the sections that specify the capability asked about. An implementation map (5720) holds, for each section and each reasoning aspect, evidence pairs consisting of a source file and a symbol that must appear in it; the pairs are checked against the program's own source files (5795) at the moment of use, the files being read as text and not executed, and the host application's self-test and measurement results (5797) are attached as further evidence. A status labeller (5730) assigns exactly one of the labels built and tested, partly built, specified and not yet built, not planned by design, and not in this application; a capability is labelled built only when all of its evidence is present. An aspect describer (5740) describes reasoning aspects — deterministic reasoning, atomic Fact Units, world model, predictive cognition, neuromorphic layer, hypothesis generation, simulation, deduction, feedback loops, supervised self-improvement, recursive self-improvement and self-knowledge — at selectable levels of detail. A consistency checker (5750) reports as a contradiction any claim labelled built whose evidence is missing, and any capability labelled specified whose implementation names are found in the program; a contradicted claim is withheld. A gap agenda (5760) lists the specified and partly built sections with their nearest built components. A rule-based answerer (5770) composes the answer with its label, section numbers and file-and-symbol sources, and a disclosure filter (5780) removes anything resembling a key or credential. Interfaces (5790) expose the summary and single answers to applications, a self-knowledge page and an administrative panel. No language model is used: the same question about the same program gives the same answer.

Reference numerals

Numeral	Element	Rests on
5700	Self-question (user, assistant or application)	FIG. 57
5705	Self-knowledge layer	FIG. 57
5710	Specification index: numbered sections §1–§56	FIG. 57
5720	Evidence-checked implementation map: (file, symbol) pairs	FIG. 57
5730	Status labeller: built / partly built / specified / not planned / not present	FIG. 57
5740	Reasoning-aspect describer: twelve aspects, three levels	FIG. 57
5750	Self-consistency checker: contradictions withheld	FIG. 57

Numeral	Element	Rests on
5760	Gap agenda: specified and partly built sections	FIG. 57
5770	Rule-based answerer: label, sections, sources	FIG. 57
5780	Disclosure filter: keys and credentials removed	FIG. 57
5790	Interfaces: /api/self, /api/self/ask, page, panel, assistant	FIG. 57
5795	The program's own source files (read as text)	FIG. 57
5797	Self-tests and measurements of the host application	FIG. 57
5799	Answer with status label and sources	FIG. 57

Status

The self-knowledge layer of FIG. 57 is built and tested in prototype v9.64 (qb_selfknow.py) and in the shared core of the applications (qbselfknow.py). It corresponds to features 383–391 of the QueryBook Composite Feature Registry v70, to Section 56 of the Core Technology Overview v15 and to claims 358–362 of the application.

Detailed Description of FIG. 58

The Digital Synaptic Network borrows the communication economics of neurons — local work, sparse signalling, thresholds, inhibition and connections that change with use — for the way QueryBook's agents find and share knowledge. It has two parts. Answer routing decides which agents (in the applications, the dataset stores of the bulk harvest) are asked for a lookup: every one in turn (Central), only those its learned pathways point to (DSN), or Central with the DSN compared beside it (Shadow). Network memory treats the relationships between agents as a layer of memory: agents that succeed together form digital synapses that strengthen with repeated success, weaken with joint failure, fade when unused and are consolidated when long proven; strongly bound agents form assemblies, and assemblies linked through their members form networks of networks.

A lookup (5800) enters an answer router (5805) whose routing setting (5810), chosen by an administrator, is central, shadow or routed. In the central setting every dataset store (5895) is asked in turn until enough facts are found. In the routed setting a pathway table (5820) holds, for each kind of lookup and each store, a pathway with a strength, a count of facts found and a count of empty answers; the stores whose pathways are at or above a threshold are asked strongest first, a store returning facts being strengthened and a store returning none weakened, weak pathways recovering slowly and one weak pathway being retried periodically. When the strong pathways return nothing, a fallback (5830) asks every remaining store, ordered by spreading activation over the network memory of FIG. 59, so that routing never withholds a fact the central setting would find. In the shadow setting the central answer is returned and the routed lookup is run on the same lookup in the background; a comparison log (5840) records agreement, facts missed and added, fallback, and for each method the facts returned, stores asked and time; and a health report (5850) states the setting in force and a verdict over the last 24 hours, no switch being recommended before a minimum number of lookups or when any fact was missed. When routing is enabled again after the central setting, a re-check (5860) moves every pathway halfway back to neutral. A stress-test harness (5870) compares central, broadcast and routed designs under data, agent, scale and load stresses.

Reference numerals

Numeral	Element	Rests on
5800	Lookup (subject text, limit)	FIG. 58
5805	Answer router	FIG. 58
5810	Routing setting: Central · Shadow · DSN (administrator)	FIG. 58
5820	Learned pathway table: kind × store → strength, facts, empty	FIG. 58
5830	Complete-answer fallback: every store, ordered by spreading activation	FIG. 58
5840	Shadow comparison log	FIG. 58
5850	Health report: routing, 24-hour comparison, verdict	FIG. 58
5860	Re-check on re-enable: pathways halfway to neutral	FIG. 58

Numeral	Element	Rests on
5870	Stress-test harness: six engines, stresses, learning over time	FIG. 58
5895	Dataset stores (agents)	FIG. 58
5899	Answer returned (Central's in Shadow)	FIG. 58

Status

The answer routing of FIG. 58 is built and tested in the shared core of the applications (qbdsn.py, Control Panel 1.34). It corresponds to features 392–398 of the QueryBook Composite Feature Registry v71, to Chapter 131 of the QueryBook Bible v2.1 (Pass 120) and to claims 363–366 and 371 of the application. The stress-test harness (5870) is a test tool, not part of the product.

Detailed Description of FIG. 59

Both parts are governed by the same rules as every QueryBook fact. Learned pathways and synapses hold relationships, never facts; they never change what an answer says; and every one of them can be inspected — its strength, its threshold, the contexts it formed in, its good and bad outcomes, its evidence and its history. The design principle is stated plainly: do not only make the agents smarter, make the connections between them able to learn — and keep those connections auditable.

Activity sources (5900) — the specialists whose facts answered one question, harvesting agents completing a cycle, the stores that answered one lookup, and a check asked by one application team and answered by another — report agents active together. A synapse store (5905) holds digital synapse records (5910), each comprising a relationship type, a strength, an activation threshold, the contexts in which it formed, counts of good and bad outcomes, recent evidence, a history of recent changes, its formation and last-use times, and a consolidation flag. A plasticity unit (5920) forms a synapse at a low strength below threshold only on a joint success, potentiates it on each joint success and depresses it on each joint failure, the threshold rising with the share of bad outcomes. A turnover unit (5930) fades every synapse according to its idle time, more slowly once consolidated by a long record of success, and removes synapses that have faded below a floor and remained idle. A co-activation unit (5940) links agents active in one episode, agents succeeding within a time window, and agents whose output another used, the last as directed links. An engram store (5950) keeps, for each context, the agents active when it was solved. An assembly detector (5960) groups agents joined by synapses above a binding strength into functional assemblies, and a link detector (5970) links assemblies through synapses above threshold between their members. A spreading-activation unit (5980) propagates activation from active agents along synapses above threshold, scaled by strength and attenuated at each step, to rank related agents. Interfaces (5990) expose the network, each synapse's explanation and a health line. Network memory records and reports; it never alters an answer. In the prototype each answer reports the assembly to which its specialists belong and the related specialists ranked by spreading activation, while its verdict, text and citations are the same as without the network.

Reference numerals

Numeral	Element	Rests on
5900	Activity sources: answers, agent cycles, store lookups, cross-application checks	FIG. 59
5905	Synapse store	FIG. 59
5910	Digital synapse record: type, strength, threshold, contexts, good/bad, evidence, history	FIG. 59
5920	Plasticity: weak formation, potentiation, depression, threshold rise	FIG. 59
5930	Turnover: fading by idle time, consolidation, pruning	FIG. 59
5940	Co-activation: episodes, time window, directed links	FIG. 59
5950	Engram store: agents active per context	FIG. 59

Numeral	Element	Rests on
5960	Assembly detector: synapses ≥ binding strength	FIG. 59
5970	Assembly links: networks of networks	FIG. 59
5980	Spreading activation: strength × falloff, above threshold	FIG. 59
5990	Interfaces: /api/network, Network page, Control Panel, health lines	FIG. 59

Status

The network memory of FIG. 59 is built and tested in prototype v9.64 (qb_synapse.py) and in the shared core of the applications (qbsynapse.py, Control Panel 1.35). It corresponds to features 399–407 of the QueryBook Composite Feature Registry v71, to Chapter 131 of the QueryBook Bible v2.1 (Pass 120) and to claims 367–374 of the application.

Detailed Description of FIG. 60

A Fact Unit is no longer read as simply held or not held. Every Fact Unit that QueryBook returns now stands in exactly one of four truth states — true, false, undetermined or conflicting — and carries, beneath that state, the backing that explains it and a weight. A fact with nothing recorded about its evidence is undetermined; it is never assumed true. A second, separate field records identity: whether the subject of the fact is resolved to one known entity, unresolved, ambiguous or conflicting. A name printed in a newspaper is true as printed and unresolved as to who.

A stored fact (6000) with its status, the sources named in its record and any contradiction mark enters a rating unit (6005). A backing selector (6020) derives one of eight backings — verified, computed, corroborated, sourced, reported, unsourced, disputed or rejected — applying a corroboration rule (6040) by which a sourced fact whose own record names two or more different sources is corroborated, and treating a contradicted fact as disputed unless it is rejected. A state table (6010) maps the backing to exactly one of four truth states: true, false, undetermined or conflicting; a fact with no recorded evidence is undetermined. A weight table (6030) attaches a fixed weight to the backing, in one embodiment 1.0 for verified and computed, 0.9 for corroborated, 0.75 for sourced, 0.4 for reported, 0.3 for disputed, 0.2 for unsourced and 0.0 for rejected. The rated fact (6050) is returned with its state, backing, weight, reason and code, and a tally of the four states accompanies the answer (6099). The rating is computed from the stored record and alters neither the record nor its identifier.

An encoder (6060) represents each state by two bits, a first bit indicating that confirming evidence exists and a second bit indicating that the assertion has been found wrong: undetermined 00, false 01, true 10, conflicting 11. A join unit (6070) combines the states of one assertion drawn from separate records by a bitwise OR of their codes, the result being independent of the order of combination and unchanged by repetition; a plurality of codes is packed into a machine word, in one embodiment 32 codes in a 64-bit word, so that a single OR operation joins the plurality. A hardware mapping (6080) places the same states on different devices: two bits per state on a binary processor; one digit per state on a device whose storage or logic element distinguishes four levels; and, on a device distinguishing eight or more levels, one digit holding the backing from which the state follows. The states correspond in meaning to the four values of hardware description — logic 1, logic 0, an undriven line and a line driven in conflict — without requiring any such circuit.

Reference numerals

Numeral	Element	Rests on
6000	Stored fact: status, sources named in its record, contradiction mark	FIG. 60
6005	Rating unit	FIG. 60
6010	State table: true · false · undetermined · conflicting	FIG. 60
6020	Backing selector: verified, computed, corroborated, sourced, reported, unsourced, disputed, rejected	FIG. 60
6030	Weight table: fixed weight per backing	FIG. 60
6040	Corroboration rule: two different sources in the fact's own record	FIG. 60

Numeral	Element	Rests on
6050	Rated fact: state, backing, weight, reason, code	FIG. 60
6060	Two-bit evidence encoder: 00, 01, 10, 11	FIG. 60
6070	Join unit: bitwise OR over packed states (32 per 64-bit word)	FIG. 60
6080	Hardware mapping: binary · four-level · eight or more levels	FIG. 60
6099	Answer with tally of the four states	FIG. 60

Status

The rating of FIG. 60 is built and tested in the shared core of the applications (sense.py; engine 1.56). The packed two-bit form (6070) is built as functions and is not yet used in the store's file format; the multi-level devices of the hardware mapping (6080) are specified, and no hardware other than an ordinary binary processor is used or required. It corresponds to features 408–415 of the QueryBook Composite Feature Registry v72, to Chapter 132 of the QueryBook Bible v2.1 (Pass 121) and to claims 375–381 of the application.

Detailed Description of FIG. 61

Whether a statement is true and who or what it is about are different questions, and each has its own four states. Identity is resolved when the subject carries its record's own identifier; unresolved when it is only a name; ambiguous when several known entities answer to it; conflicting when identifiers disagree. A name harvested from a newspaper page is recorded as printed — true as printed, status reported — and its identity is unresolved: it stands for whoever bore that name. It is shown apart from any known person of the same name and is never merged by name alone.

An identity unit (6110) assigns to the subject of a fact, apart from the truth state, one of four identity states: resolved, when the subject carries its record's own identifier; unresolved; ambiguous; or conflicting. An as-printed fact (6160) records a name printed in a publication together with the date, publication, place, page, a reason and the publication's own words; its truth state is that of a report and its identity state is unresolved, and it is presented in an unresolved zone (6120) apart from a known entity of the same name, a name alone never merging two records. A mention weigher (6130) compares each such mention with a known entity's own facts — dates of birth and death, offices, places — and reports likely, possible or impossible with reasons, as an analysis only. A context table (6140) holds, beside each fact and keyed by item, predicate and stored value, the statement's start, end and point in time, qualifiers, rank and sources, the fact and its identifier being unchanged; a fill unit (6150) reads a subject's record from the source its facts cite and adds such context on request, never adding a fact and never supplying a date the source does not state. A separation (6170) keeps observed measurements, observed navigation and inferred intent apart from facts, so that demand orders harvesting and never changes a fact's state or weight.

Reference numerals

Numeral	Element	Rests on
6100	Subject of a fact	FIG. 61
6105	Identity and context layer	FIG. 61
6110	Identity unit: resolved · unresolved · ambiguous · conflicting	FIG. 61
6120	Unresolved zone: shown apart, never merged by name	FIG. 61
6130	Mention weigher: likely · possible · impossible, with reasons	FIG. 61
6140	Context table beside the facts: start, end, point in time, qualifiers, rank, sources	FIG. 61
6150	Fill unit: reads the cited source; adds context only	FIG. 61
6160	As-printed fact: name, date, publication, page, reason, words	FIG. 61
6170	Separation: measurements, navigation and inferred intent apart from facts	FIG. 61
6199	Answer: known entity, mention analysis, unresolved zone	FIG. 61

Status

The identity state, unresolved zone, mention weigher, context table and fill unit of FIG. 61 are built and tested in the shared core of the applications (sense.py, qbbulk.py, qbnews.py; engine 1.56); the ambiguous and conflicting identity states are defined and not yet assigned automatically. It corresponds to features 416–422 of the QueryBook Composite Feature Registry v72, to Chapter 132 of the QueryBook Bible v2.1 (Pass 121) and to claims 382–386 of the application.